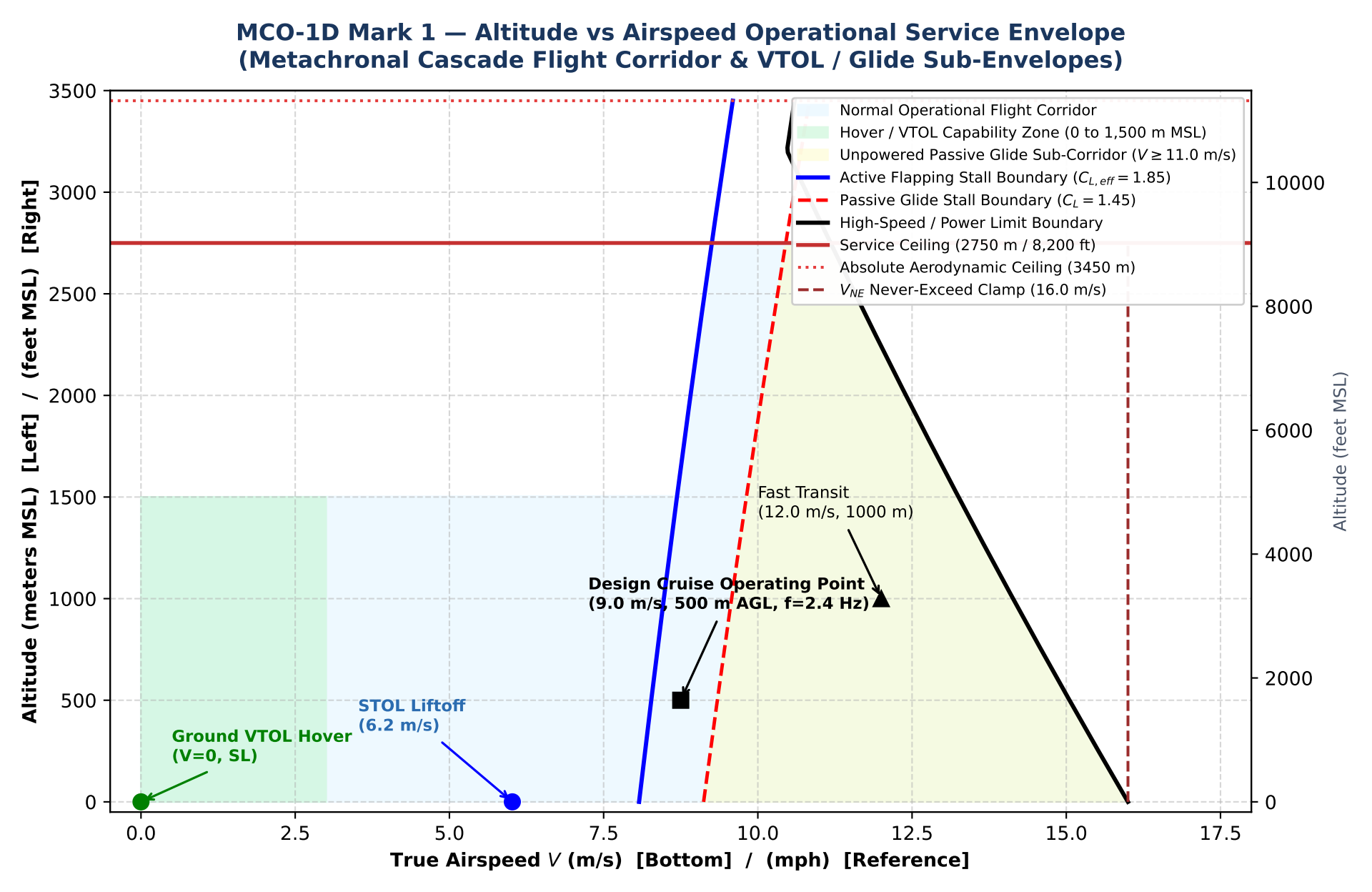
MCO-1D Mark 1 — Altitude vs Airspeed Operational Service Envelope
(Metachronal Cascade Flight Corridor & VTOL / Glide Sub-Envelopes)
Normal Operational Flight Corridor
Hover / VTOL Capability Zone (0 to 1,500 m MSL)
Unpowered Passive Glide Sub-Corridor ($V \geq 11.0$ m/s)
Active Flapping Stall Boundary ($C_{L,eff} = 1.85$)
Passive Glide Stall Boundary ($C_L = 1.45$)
High-Speed / Power Limit Boundary
Service Ceiling (2750 m / 8,200 ft)
Absolute Aerodynamic Ceiling (3450 m)
V_{NE} Never-Exceed Clamp (16.0 m/s)
Fast Transit
(12.0 m/s, 1000 m)
Design Cruise Operating Point
(9.0 m/s, 500 m AGL, f=2.4 Hz)
STOL Liftoff
(6.2 m/s)
Ground VTOL Hover
(V=0, SL)
Altitude (meters MSL) [Left] / (feet MSL) [Right]
True Airspeed V (m/s) [Bottom] / (mph) [Reference]
Altitude (feet MSL)
0
500
1000
1500
2000
2500
3000
3500
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
2000
4000
6000
8000
10000